

TOMAHAWK AGDM, AB, Canada

WMO: 715660

Lat: 53.4395N

Lon: 114.7189W

Elev: 2671

StdP: 13.33

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.9	-14.2	-26.1	1.4	-19.4	-20.4	2.0	-13.5	20.8	32.6	18.4	26.9	4.2	0	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	81.2	62.3	78.0	60.9	75.2	59.6	65.4	76.9	63.3	74.6	61.5	72.1	6.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	89.0	71.0	58.9	82.1	68.2	57.0	76.4	66.1	31.7	76.2	30.1	74.1	28.7	72.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.6	15.9	13.7	DB	-27.3	87.9	5.5	4.2	-31.3	90.9	-34.5	93.4	-37.6	95.7	-41.6	98.7
			WB	-27.5	68.7	5.0	2.6	-31.1	70.6	-34.0	72.1	-36.8	73.6	-40.5	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	17.7	16.5	26.7	39.1	51.0	56.8	61.8	59.2	51.2	39.9	25.5	14.9
	DBStd	19.73	14.66	14.34	12.56	9.00	7.66	5.21	5.29	5.67	7.84	8.02	12.38	13.62
	HDD50	5316	1001	938	722	339	80	7	0	4	77	324	735	1089
	HDD65	9724	1466	1358	1187	778	435	249	129	192	414	778	1184	1554
	CDD50	1119	0	0	0	12	112	213	365	290	115	11	1	0
	CDD65	51	0	0	0	0	2	5	29	13	2	0	0	0
	CDH74	885	0	0	0	3	75	89	392	225	98	3	0	0
	CDH80	147	0	0	0	0	7	9	78	39	14	0	0	0
Wind	WSAvg	6.1	5.6	5.8	6.3	7.3	7.2	6.5	5.9	5.5	5.7	6.0	5.5	5.4
Precipitation	PrecAvg	19.3	1.0	0.6	0.7	1.2	2.2	3.4	3.7	2.5	1.5	0.9	0.8	0.7
	PrecMax	25.7	3.0	1.2	1.4	2.2	4.3	6.9	8.2	5.4	3.4	3.0	2.4	1.9
	PrecMin	12.6	0.1	0.0	0.1	0.2	0.7	1.0	1.8	1.2	0.4	0.2	0.2	0.2
	PrecStd	3.1	0.7	0.3	0.3	0.6	0.9	1.3	1.5	0.8	0.9	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	48.6	56.4	71.7	80.6	81.3	86.4	84.2	81.5	70.3	57.8	46.5
		MCWB	38.1	37.3	42.1	51.2	55.2	62.0	66.0	63.6	61.4	52.8	44.4	37.7
	2%	DB	44.1	42.7	50.9	64.7	75.6	76.5	81.7	79.4	76.6	61.5	49.0	40.7
		MCWB	36.3	34.5	39.4	47.0	54.0	59.6	64.3	62.7	58.9	48.1	40.0	33.6
	5%	DB	40.0	39.1	46.4	59.0	71.3	72.8	78.2	76.1	71.6	57.0	44.0	36.8
		MCWB	33.5	32.2	36.7	43.7	52.2	57.6	62.7	61.9	56.8	45.3	37.0	31.2
	10%	DB	36.2	35.5	42.5	54.4	66.6	69.4	75.0	72.6	65.7	52.8	39.9	32.7
		MCWB	31.3	30.0	34.5	41.6	50.0	56.2	61.1	60.2	53.8	43.0	34.1	28.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	38.2	42.8	51.2	57.9	64.1	69.2	67.6	62.7	53.7	44.9	37.8
		MCDB	47.8	46.9	54.0	71.2	75.4	77.3	79.8	77.9	77.6	68.0	55.7	44.9
	2%	WB	36.5	35.1	39.8	47.3	55.3	61.3	66.5	64.8	59.8	49.0	40.8	34.5
		MCDB	43.3	42.0	50.0	62.8	70.4	73.3	77.8	76.2	74.6	59.6	47.6	40.0
	5%	WB	34.1	32.7	37.2	44.6	53.3	59.2	64.1	62.9	57.2	46.3	37.8	31.7
		MCDB	39.4	38.4	45.9	57.3	68.0	70.2	75.4	73.4	68.8	55.2	43.4	36.2
	10%	WB	31.5	30.3	34.8	42.2	51.3	57.4	62.1	61.0	54.8	43.9	34.8	28.8
		MCDB	36.1	35.4	41.6	52.7	64.9	66.8	72.7	70.9	64.7	51.7	39.0	32.7
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.3	18.5	19.4	22.1	20.4	22.1	22.0	22.2	18.5	14.7	14.8
		MCDBR	19.5	20.6	23.7	28.3	29.8	26.1	27.3	28.2	32.5	26.7	20.2	17.3
		MCWBR	13.6	14.0	14.2	14.5	13.3	12.2	12.7	13.7	15.7	14.9	12.9	12.5
	5% WB	MCDBR	19.4	20.7	22.2	25.9	26.8	22.9	25.0	25.8	29.5	25.3	19.4	17.1
		MCWBR	14.1	14.5	13.4	13.7	12.4	11.6	12.8	13.6	14.7	14.6	13.0	12.8
Clear-Sky Solar Irradiance	taub	0.245	0.258	0.290	0.328	0.343	0.352	0.351	0.357	0.313	0.281	0.254	0.238	
	taud	2.366	2.407	2.420	2.390	2.392	2.382	2.440	2.376	2.500	2.571	2.473	2.333	
	Ebn at Noon	242	276	287	287	287	285	283	274	276	266	239	220	
	Edh at Noon	17	23	28	33	35	36	33	33	26	19	16	15	
All-Sky Solar Radiation	RadAvg	304	627	1092	1469	1762	1879	1921	1589	1103	634	317	220	
	RadStd	23	48	69	133	128	127	117	108	113	67	29	19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page